ZOO501P – DEVELOPMENTAL BIOLOGY

Practical Exam Notes

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M.C.Qs (Multiple Choice Questions)

1. Which of the following method is preferred for defense security clearness? **FINGERPRINT**
2. During induced spawning brood fish in control group are injected with **0.9% NaCl**.
3. If egg is **OPAQUE & BLACK**, it probably contain unhealthy embryo.
4. Infectious stage of *trichostrongylus* is **L3**.
5. In an assault case, the predator may have left the fingerprint on **VICTIM SKIN**.
6. The **ZONA PELLUCIDA** is glycoprotein matrix which acts as barrier to sperm entry in mammals.
7. Fertilization is an **ACTIVE** process.
8. There are **THREE** types of germ layers.
9. There are **EIGHT** glycoproteins.
10. Gastrulation gives rise to **GERM LAYERS**.
11. Sea Urchin exhibits **RADIAL** holoblastic cleavage.
12. Sperm was discovered in **1678**.
13. There are **17000** pores in eggshell.
14. Group of protostomes is called **LOPHOTROCHOZOA**.
15. Metamorphosis is regulated by **THYROID GLANDS**.
16. Cadherin refers to **CELL TO CELL ADHESION**.
17. *Rana catesbiana* can lay as much as **20000** eggs.
18. In mammals, polyspermy is controlled by **SLOW BLOCK**.
19. In general, polyspermy is controlled by **FAST BLOCK**.
20. In floatation technique, parasitic egg and cyst **RAISE TO THE TOP**.
21. How many *trichostrongylus* species have been associated with humans? **10**
22. What is the mode of transmission of trichostrongylus larval stage? **WATER**
23. Dactylography works on the basis of **REFERENCE POINT** on finger prints.
24. **BLASTULA** is a structure consist of inner cell mass and trophoblast in mammals.

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25. The nucleus of hen egg is located in **BLASTODISC**.
26. The male gamete only contributes its **HAPLOID NUCLEUS** to zygote during the process of fertilization.
27. Male gamete is also called **SPERMATOZOA**.
28. **KARMER** defined the chick egg quality.
29. External quality is focused on **SHELL CLEANLINESS**.
30. **BLASTOMERE** are individual cells that are produced as the result of the cleavage.
31. During oogenesis, mother deposits **mRNA** which takes over after fertilization after to orchestrate the earliest stage of embryogenesis.
32. In axolotl salamander, the cleavage furrow extends through the animal hemisphere at the rate of about **1 MM PER MINUTE.**
33. The technique where fish is stimulated for ovulation is called **INDUCED SPAWNING**.
34. During induced spawning, brood fish in the control group are injected with a **PLACEBO (USUALLY SALINE SOLUTION OR DISTILLED WATER).**
35. The time interval from the first injection to ovulation is called the **LATENCY PERIOD**.
36. The male gamete is **SMALL AND MOTILE**.
37. Chalazae are twisted in opposite directions and serve to keep the yolk **CENTERED**.

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SHORT QUESTIONS

1. Write role of acrosome cap in fertilization.

The acrosome cap is a specialized structure located at the anterior part of the sperm head and plays a crucial role in fertilization. It contains a variety of hydrolytic enzymes such as hyaluronidase, acrosin, and proteases that are essential for penetrating the outer layers of the ovum.

During fertilization, when the sperm reaches the egg, it undergoes the acrosome reaction. In this process, the acrosomal membrane fuses with the sperm plasma membrane, releasing enzymes from the acrosome cap. These enzymes help digest the zona pellucida (in mammals) or egg envelopes in other organisms, allowing the sperm to create a passage through which it can reach the oocyte membrane.

In addition to enzymatic digestion, the acrosome cap also assists in species-specific recognition and binding between sperm and egg. It ensures that only compatible sperm can attach to and penetrate the egg, preventing fertilization by sperm of other species. After successful penetration, the sperm membrane fuses with the egg membrane, enabling the transfer of genetic material and initiating fertilization.

2. Differentiate between zygote and ovum.

OVUM

The ovum is the mature female gamete (egg cell) produced by the ovary through oogenesis. It is haploid (n), meaning it contains a single set of chromosomes. The ovum is specialized for receiving the male gamete and does not undergo further division unless fertilization occurs. It is metabolically active and contains nutrients and cytoplasmic materials needed for early development after fertilization.

ZYGOTE

The zygote is the first diploid (2n) cell formed immediately after the fusion of the male gamete (sperm) and the ovum during fertilization. It contains a complete set of chromosomes from both parents. Unlike the ovum, the zygote is the starting point of embryonic development and undergoes rapid mitotic divisions to form a multicellular organism.

3. How *trichostrongylus* effects human?

In humans, *Trichostrongylus* infection primarily affects the small intestine. After ingestion, the larvae develop into adult worms that attach to the intestinal mucosa. This leads to irritation and mild damage of the intestinal lining, interfering with normal digestion and nutrient absorption.

Clinically, infection may cause abdominal pain, diarrhea, nausea, loss of appetite, and weight loss. In some cases, patients may experience weakness and mild anemia due to chronic intestinal irritation and nutrient depletion. Heavy infections are uncommon in humans but can lead to more pronounced gastrointestinal symptoms.

Overall, *Trichostrongylus* affects humans by causing intestinal helminth infection that results in digestive disturbances and nutritional impairment, especially in areas where sanitation is poor or raw contaminated vegetables are consumed.

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4. What is ovaprim and why it is used in fish breeding?

OVAPRIM

Ovaprim is a synthetic hormone preparation widely used in aquaculture to induce breeding in fish. It contains a combination of a gonadotropin-releasing hormone (GnRH) analog and a dopamine antagonist, which together stimulate the reproductive system of fish.

In normal conditions, fish reproduction is controlled by hormones from the brain and pituitary gland. However, in captivity, environmental conditions are often not sufficient to trigger natural spawning. Ovaprim bypasses this limitation by directly stimulating the release of gonadotropins from the pituitary, which then promote maturation of gametes and induce ovulation in females and spermiation in males.

It is used in fish breeding because it provides a reliable and controlled method for inducing spawning, especially in commercially important species. It improves breeding success, ensures synchronization of spawning, and allows production of fish seed in hatcheries throughout the year, independent of natural breeding seasons.

5. What will happen if we wash chicken hen manually in water?

If a chicken hen is washed manually with water, it can have several harmful effects on its health and normal functioning.

- ✓ **First**, washing removes the natural oil layer present on the feathers. This oil, produced by the uropygial (preen) gland, helps maintain waterproofing and insulation. Without it, the bird may lose heat more quickly and become more vulnerable to cold stress and chilling.
- ✓ **Second**, wetting the feathers can disrupt their structure and reduce their ability to trap air, which normally helps in temperature regulation and buoyancy. This can make the hen uncomfortable and stressed.
- ✓ **Third**, manual washing can increase the risk of skin irritation and may expose the bird to infections if proper drying and care are not ensured. Birds generally maintain hygiene through preening behavior rather than external washing.

6. Differentiate between acrosomal granules and cortical granules.

ACROSOMAL GRANULES

Acrosomal granules are found in the head of the sperm cell within the acrosome, a cap-like structure derived from the Golgi apparatus during spermatogenesis. These granules contain hydrolytic enzymes such as acrosin and hyaluronidase. During fertilization, the acrosomal granules are released through the acrosome reaction when the sperm contacts the egg. Their main function is to help the sperm penetrate the outer layers of the egg, especially the zona pellucida or egg envelope.

CORTICAL GRANULES

Cortical granules, on the other hand, are present just beneath the plasma membrane of the ovum (egg cell) in the cortex of the egg cytoplasm. They are formed during oogenesis. After the sperm enters the egg, cortical granules are released in a process called cortical reaction. Their enzymes modify the zona pellucida and prevent entry of additional sperm, thereby ensuring polyspermy block.

7. Write importance of vitelline membrane.

Main importance of Vitelline membrane is that it provides structural support and protection to the egg cell, helping to maintain its shape and integrity. It also acts as a selective barrier that regulates the entry of sperm and other substances, ensuring controlled interaction during fertilization. During fertilization, the vitelline membrane plays a key role in sperm recognition and binding. It contains specific receptor sites that allow only compatible sperm to attach, contributing to species-specific fertilization. After sperm entry, the vitelline membrane undergoes changes (such as elevation or hardening in many species), forming a fertilization membrane that prevents polyspermy, which is the entry of multiple sperm into the egg.

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8. Write method used for examination of parasitic fecal sample.

The examination of a parasitic fecal sample is mainly done to detect the presence of helminth eggs, larvae, or protozoan cysts. Several methods are used depending on the type and intensity of infection.

- ✓ The most common method is the direct wet mount examination. In this method, a small amount of fresh stool is mixed with saline or iodine solution on a glass slide and covered with a cover slip. It is then observed under a microscope to identify parasite eggs, cysts, or motile trophozoites. This method is simple and quick but less sensitive for light infections.
- ✓ Another important method is the concentration technique, which increases the chances of detecting parasites by separating them from fecal debris. It includes sedimentation and flotation methods. In sedimentation, parasites settle at the bottom after centrifugation, while in flotation, lighter parasite eggs float on a special solution and are collected for microscopic examination.
- ✓ A third method is the Kato-Katz thick smear technique, which is commonly used for quantitative estimation of helminth eggs in stool, especially in field surveys.

9. What is nucleus if pander? Write its significance.

NUCLEUS OF PANDER

The nucleus of Pander (also called the germinal disc nucleus) is a specialized nucleus present in the animal pole region of the ovum of birds and reptiles. It is named after the embryologist Heinz Christian Pander, who studied early embryonic development in egg-laying animals. This nucleus is located within the germinal disc, which is a small, disc-shaped area of cytoplasm in the egg where active development takes place after fertilization. It is the region that contains the genetic material that will participate in early cleavage and embryo formation.

Significance of nucleus of Pander:

It is the site where fertilization and early embryonic development are initiated in birds and reptiles. After the sperm enters the egg, the nucleus of Pander participates in the formation of the zygote and supports rapid cell division (cleavage). It plays a key role in organizing the early developmental processes, leading to the formation of the embryo.

10. Why corona radiata is important during fertilization?

CORONA RADIATA

The corona radiata is the outermost layer of follicular cells surrounding the ovum (egg). It plays several important roles during fertilization.

Importance of Corona Radiata

Its main importance is that it provides protection and nourishment to the egg cell before fertilization. These surrounding cells help maintain the viability of the ovum as it travels through the oviduct. During fertilization, the corona radiata acts as the first barrier for sperm entry. Sperm must pass through this layer before reaching the zona pellucida and the egg membrane. To do this, sperm release enzymes such as hyaluronidase, which help disperse the follicular cells of the corona radiata. It also contributes to sperm selection and guidance, allowing only healthy and motile sperm to move closer to the egg. This increases the chances of successful fertilization.

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11. How fingerprinting is important in Amnesia and Deceased cases?

In amnesia cases, where a person has lost memory and cannot recall their identity, fingerprinting helps in establishing identity by matching their prints with existing records in national databases. Once matched, authorities can retrieve personal information such as name, address, family details, and medical history, helping in reunification and legal identification.

In deceased cases, especially when bodies are unrecognizable due to decomposition, accidents, burns, or disasters, fingerprinting is used for identification. The fingerprints of the deceased are collected and compared with stored records. Since fingerprints are permanent and highly specific to each individual, they provide reliable proof of identity even when other features are destroyed.

12. Define air cell and chalaza.

AIR CELL

The air cell is a small pocket of air located between the inner and outer shell membranes at the broader end of the egg. It forms after the egg is laid due to cooling and contraction of the egg contents. Its main function is to provide oxygen for the developing embryo during later stages of incubation.

CHALAZA

The chalaza is a pair of twisted, rope-like structures made of albumen (egg white). They extend from both sides of the yolk and attach it to the shell membranes. The chalaza keeps the yolk suspended in the center of the egg and maintains the correct orientation of the embryo during development.

13. What happens when mitosis occurs during gamete formation?

If mitosis occurs during gamete formation, it disrupts the normal process of reproduction. Normally, gametes (sperm and egg cells) are formed by meiosis, which reduces the chromosome number to half (haploid, n). This ensures that when fertilization occurs, the zygote restores the normal diploid (2n) chromosome number. If mitosis takes place instead of meiosis during gamete formation, the chromosome number is not reduced. As a result, the gametes remain diploid (2n) rather than haploid. When such gametes fuse during fertilization, the resulting zygote will have an abnormally doubled chromosome number (4n or higher). This leads to chromosomal imbalance (polyploidy or genetic abnormalities), which usually causes abnormal development, infertility, or non-viable offspring.

14. Write functions of chalaza.

Functions of chalaza:

- ***Anchors the yolk in position:*** The chalaza consists of twisted strands of thick albumen that attach the yolk to both ends of the egg, keeping it suspended in the center and preventing it from shifting freely.
- ***Maintains correct orientation of embryo:*** It helps keep the germinal disc (where the embryo develops) in the proper upper position, which is important for normal embryonic development.
- ***Provides stability inside the egg:*** By holding the yolk in place, the chalaza ensures that the internal structure of the egg remains stable even when the egg is moved or shaken.
- ***Acts as a protective cushion:*** It reduces sudden movement and shock to the yolk, protecting the developing embryo from mechanical damage.
- ***Supports proper development conditions:*** By maintaining yolk position, it ensures efficient nutrient utilization and proper spatial arrangement for early embryo growth.

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15. Define fertilization in mammals.

Fertilization in mammals:

- Fertilization is the biological process in which a male gamete (sperm) fuses with a female gamete (ovum/egg) to form a diploid zygote.
- In mammals, it usually occurs in the ampulla region of the fallopian tube (oviduct).
- During this process, the sperm penetrates the ovum, and their nuclei fuse, combining genetic material from both parents.
- This event restores the diploid chromosome number (2n) and initiates embryonic development.
- Fertilization also determines the sex of the offspring based on the type of sperm (X or Y chromosome) involved.

16. Write function of mitochondria in sperm cell.

Function of mitochondria in sperm cell:

- ***Produces energy (ATP):*** Mitochondria in sperm are responsible for generating ATP through cellular respiration, which provides the energy needed for sperm movement.

- ***Enables motility:*** The ATP produced powers the beating of the flagellum (tail), allowing the sperm to swim through the female reproductive tract toward the egg.

- ***Located in midpiece:*** Mitochondria are tightly packed in the middle piece of the sperm, forming a sheath around the flagellum to supply continuous energy.

- ***Supports fertilization process:*** By ensuring sustained movement, mitochondria indirectly increase the chances of the sperm reaching and penetrating the ovum successfully.

17. What are dosage of Ovaprim for male and female fishes?

Dosage of Ovaprim for male and female fishes:

- ***Female fishes:***
  - Usually 0.5 ml/kg to 1.0 ml/kg body weight
  - Common practice is 0.5 ml/kg for initial dose and 1.0 ml/kg for final (resolving) dose depending on species and maturity

- ***Male fishes:***
  - Generally lower dose than females, about 0.1 ml/kg to 0.5 ml/kg body weight
  - Often a single low dose is enough to induce spermiation

18. Differentiate between germ layer and blastomere.

GERM LAYER

A germ layer refers to one of the primary layers of cells formed during embryonic development after gastrulation. These layers are the ectoderm, mesoderm, and endoderm, and each is responsible for forming specific tissues and organs of the body. Germ layers are organized and differentiated structures that contribute directly to the formation of the body plan and organ systems.

BLASTOMERE

A blastomere, on the other hand, is an individual cell formed during the early cleavage divisions of the zygote. These cells are produced by repeated mitotic divisions without growth in size. Blastomeres are undifferentiated cells and serve as the basic units that later give rise to the blastula and eventually contribute to the formation of germ layers during gastrulation.

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19. How will you identify unfertilized egg?

An unfertilized egg can be identified by its external appearance, internal structure, and lack of embryonic development. In most animals and birds, an unfertilized egg shows no signs of cleavage or cell division. The germinal disc remains small, clear, and inactive because no sperm has entered the ovum to initiate development. There is also no formation of embryo or blastoderm, which are normally seen in fertilized eggs after incubation.

In birds, when a freshly laid egg is cracked open, an unfertilized egg shows a simple, uniform germinal spot on the yolk, whereas a fertilized egg shows a bull's-eye appearance (blastodisc with concentric rings**)** due to early cell division. Additionally, unfertilized eggs do not develop further even if kept under incubation conditions. They may eventually deteriorate without any embryonic growth.

20. Which technique is used to separate parasites from fecal sample?

The technique used to separate parasites from a fecal sample is called the concentration technique. This method is used to increase the chances of detecting parasite eggs, cysts, or larvae by separating them from fecal debris. It is mainly of two types:

- ✓ The sedimentation method, in which parasite elements are heavier and settle at the bottom after mixing and centrifugation.
- ✓ The flotation method, in which a special solution (such as salt or sugar solution) is used so that parasite eggs float to the surface and can be collected for microscopic examination.

21. What is the function of subgerminal layers in avian eggs?

The subgerminal layer in avian eggs is a thin layer of fluid or cytoplasmic space located just beneath the blastodisc (germinal disc) on the surface of the yolk.

Function of subgerminal layer:

- It helps in separating the blastodisc from the yolk, creating a clear area for early embryonic development.
- It provides a supporting medium for cell division and expansion during early cleavage stages.
- It plays a role in facilitating diffusion of nutrients from the yolk to the developing embryo.
- It contributes to the formation of the blastoderm, which later gives rise to the embryo.
- It ensures proper structural organization of early embryonic cells in birds.

22. What will happen if neural tube does not fuse?

If the neural tube does not fuse properly during embryonic development, it leads to neural tube defects (NTDs), which are serious congenital abnormalities of the brain and spinal cord. When the anterior (head) part of the neural tube fails to close, it results in anencephaly, where major parts of the brain and skull do not develop properly. This condition is usually fatal either before birth or shortly after. If the posterior (spinal) part of the neural tube fails to close, it leads to spina bifida, where the spinal cord and vertebrae do not form completely. This can cause physical disabilities such as paralysis, loss of sensation in the lower limbs, and bladder or bowel dysfunction depending on severity.

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23. What role does axoneme plays in the in human sperm?

The axoneme is the central structural component of the sperm flagellum (tail) and plays a crucial role in sperm movement. It consists of a characteristic "9 + 2" arrangement of microtubules—nine pairs of outer microtubules surrounding two central microtubules. This structure is supported by motor proteins such as dynein, which generate sliding movements between microtubules.

The main role of the axoneme in human sperm is to produce motility. The coordinated bending and sliding of microtubules generate the whip-like movement of the tail, allowing the sperm to swim through the female reproductive tract toward the ovum. It also ensures directional movement and propulsion, which is essential for the sperm to reach the egg and achieve fertilization. Without a functional axoneme, sperm would be immotile and unable to fertilize the egg.

24. What will happen if the sperm lacks centriole?

If the sperm lacks a centriole, fertilization and early embryonic development are severely affected. The centriole in sperm is essential because it organizes the microtubules required for cell division after fertilization. Normally, after the sperm enters the ovum, its centriole helps in forming the centrosome, which is responsible for organizing the mitotic spindle. If the sperm lacks a centriole:

- The zygote cannot properly form the spindle apparatus needed for mitotic division.
- This leads to failure of normal cleavage (cell division) in the early embryo.
- As a result, embryonic development stops or becomes abnormal, leading to infertility or early embryonic death.

LONG QUESTIONS

1. Write role of finger printing in different field.

Role of fingerprinting in different fields:

- ***Forensic / Criminal investigation:*** Fingerprinting is widely used in criminal investigations to identify suspects by matching fingerprints collected from crime scenes with existing records. It is highly reliable because fingerprints are unique to every individual. It helps in solving serious crimes such as theft, murder, and assault. Courts also accept fingerprint evidence as strong legal proof. In many cases, it is used to link a criminal to multiple crime scenes.

- ***Civil identification:*** In civil administration, fingerprinting is used for personal identification in national identity cards, passports, voter registration, and driving licenses. It ensures that each individual has a unique identity in government records. It also helps prevent identity fraud and duplication. Many organizations also use fingerprint systems for attendance and employee verification.

- ***Banking and financial security:*** Fingerprinting is used in banking systems for secure authentication of customers. It is commonly used in ATMs, mobile banking apps, and online transactions. This prevents unauthorized access to accounts and reduces financial fraud. It is also widely used in smartphones and digital payment systems for secure login and verification.

- ***Immigration, border control, and medical use:*** Fingerprinting is used at airports and border checkpoints to verify traveler identities and maintain security records. It helps in tracking movement of individuals across countries. In medical fields, fingerprinting is used to identify unconscious or unknown patients. It ensures correct patient records and prevents identification errors in hospitals.

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2. What do you know about the structure of a hen egg?

STRUCTURE OF A HEN EGG:

Shell: The shell is the outermost hard, calcareous covering of the egg. It is porous in nature, allowing exchange of gases like oxygen and carbon dioxide. It also protects the internal contents from mechanical injury and microbial infection.

Shell membranes: Just beneath the shell are two thin membranes known as the outer and inner shell membranes. They act as protective barriers against bacteria and help in maintaining the shape of the egg.

Air cell: The air cell is a small pocket of air formed at the broader end of the egg between the two shell membranes. It develops after laying due to cooling and provides oxygen to the developing embryo during incubation.

Albumen (egg white): The albumen is a clear, protein-rich fluid surrounding the yolk. It provides water, proteins, and protection to the embryo. It also acts as a shock absorber and is divided into thick and thin layers.

Chalazae: Chalazae are twisted, rope-like strands of albumen that extend from both sides of the yolk. They hold the yolk in the center of the egg and maintain its proper position and orientation.

Yolk: The yolk is the main nutritive part of the egg, rich in fats, proteins, vitamins, and minerals. It serves as the primary food source for the developing embryo.

Vitelline membrane: The vitelline membrane surrounds the yolk and keeps it intact. It also plays a role in sperm binding during fertilization and helps maintain the structure of the yolk.

Germinal disc: The germinal disc is a small whitish spot on the surface of the yolk. It is the site where fertilization occurs and embryonic development begins.

3. Differentiate between white yolk and yellow yolk.

Difference between White yolk and Yellow yolk

Feature	White yolk (Latebra / central yolk)	Yellow yolk
Location	Present at the central region of the yolk	Surrounds the white yolk in concentric layers
Colour	Whitish or pale in appearance	Yellow due to presence of pigments (carotenoids)
Formation	Formed first during early stages of egg development	Formed later and deposited around white yolk
Composition	Rich in proteins, relatively low in fats and pigments	Rich in lipids (fats), proteins, vitamins, and pigments
Quantity	Present in small amount	Present in larger quantity, forms major part of yolk
Function	Provides early nutrition during initial embryonic stages	Acts as main long-term nutrient reserve for embryo development
Structure	Found in concentric layers within yolk	Forms outer mass surrounding white yolk
Nutritional role	Supplies basic proteins for early metabolism	Supplies energy, growth nutrients, and vitamins for full development

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4. Differentiate between inner shell and outer shell of chicken egg.

Difference between Inner shell membrane and Outer shell membrane of chicken egg

Feature	Outer shell membrane	Inner shell membrane
Position	Lies just beneath the calcareous eggshell	Lies just inside the outer shell membrane, surrounding the albumen
Location in egg	Closest to the shell	Closer to the albumen (egg white)
Thickness	Relatively thicker than inner membrane	Slightly thinner
Function	Provides first protective barrier against external injury and microbes	Acts as a selective barrier and supports formation of air cell
Role in air cell formation	Not directly involved	Helps in formation of air cell at the broad end after laying
Permeability	Semi-permeable	More selectively permeable
Protection	Prevents mechanical damage and microbial entry	Prevents bacterial invasion into albumen and maintains internal environment

5. Define terms of cleavage pattern and amount of yolk difference between spiral holoblastic, discoidal and meroblastic.

Spiral holoblastic cleavage and yolk amount: Spiral holoblastic cleavage is a type of complete (holoblastic) cleavage in which the entire egg divides, and the blastomeres are arranged in a spiral pattern. This type of cleavage occurs in eggs that contain a low to moderate amount of yolk (mesolecithal eggs). Because yolk is limited, it does not obstruct cell division, allowing uniform division of the whole egg. It is commonly seen in annelids and molluscs.

Discoidal cleavage and yolk amount: Discoidal cleavage is a type of incomplete (meroblastic) cleavage in which cell division is restricted to a small disc of cytoplasm called the blastodisc present on the surface of a large yolk. It occurs in eggs with a very high amount of yolk (telolecithal eggs), such as those of birds and reptiles. Due to the heavy yolk, cleavage cannot occur throughout the egg and is limited to the animal pole region.

Meroblastic cleavage and yolk amount: Meroblastic cleavage refers to incomplete cleavage where only a portion of the egg divides because of the presence of a large amount of yolk. In such eggs, the yolk interferes with complete division, so cleavage occurs either in a disc (discoidal type) or superficially. It is typical of telolecithal eggs found in birds, reptiles, and some insects, where yolk is abundant and restricts full cell division.

6. What observations you find while looking at nematodes?

When nematodes are observed under a microscope or in preserved samples, they show a long, slender, cylindrical, and unsegmented body. The body is bilaterally symmetrical and tapers at both ends, giving a thread-like appearance. They are covered by a tough, flexible cuticle which protects them and helps maintain body shape.

Internally, a complete digestive tract is visible with a distinct mouth at the anterior end and an anus at the posterior end. This is one of the key identifying features of nematodes. Movement is usually observed as a characteristic whip-like or serpentine motion due to longitudinal muscles only.

Sexual dimorphism may also be observed in many species, where males are smaller and often have curved posterior ends, while females are larger with straighter bodies. Reproductive structures may be visible in mature specimens depending on the species.

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7. What is dactylography? Why it is significant in identification?

DACTYLOGRAPHY:

Dactylography is the branch of forensic science that deals with the study, recording, classification, and comparison of fingerprints. It is based on the analysis of ridge patterns present on the fingertips, palms, and soles. These ridge patterns are formed during fetal development and remain unchanged throughout a person's life, making them highly reliable for identification.

Significance in identification:

Dactylography is extremely important in identification because fingerprints are unique to every individual, even identical twins do not share the same pattern. Since fingerprints are permanent and cannot be altered naturally, they provide a stable means of identification throughout a person's life. This makes them very useful in forensic investigations, where latent fingerprints collected from crime scenes are matched with criminal databases to identify suspects and solve cases.

In civil applications, dactylography is used in national identity systems, passports, voter registration, and driving licenses to ensure accurate identification and prevent identity fraud. It is also widely used in modern biometric systems such as mobile phone unlocking, banking security, and attendance systems.

8. Differentiate between Nucleus of pander and germinal disc.

Difference between Nucleus of Pander and Germinal Disc:

Feature	Nucleus of Pander	Germinal Disc
Definition	Nuclear region associated with yolk organization in avian egg	Small disc-like area on yolk where embryo develops
Location	Present within the yolk cytoplasm	Present on the surface of yolk at animal pole
Structure	Not a distinct visible disc; internal nuclear region	Clearly visible whitish circular disc
Role	Involved in early organization and regulation of egg contents	Site of fertilization and early embryonic development
Developmental importance	Indirect role in early development processes	Direct role in cleavage and formation of blastoderm
Visibility	Not easily visible under simple observation	Easily visible as a white spot on yolk
Outcome	Supports early developmental environment	Gives rise to embryo after fertilization

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9. Explain the structure of sperm.

STRUCTURE OF SPERM:

Head: The head is the anterior part of the sperm and contains the nucleus, which carries haploid genetic material (n = 23 chromosomes in humans). It is covered by the acrosome at its tip. The acrosome is a cap-like structure derived from the Golgi apparatus and contains enzymes such as hyaluronidase and acrosin, which help in penetrating the egg coverings during fertilization.

Middle piece: The middle piece lies just behind the head and contains numerous mitochondria arranged in a spiral manner. These mitochondria produce ATP through cellular respiration, which provides energy required for sperm movement.

Tail (Flagellum): The tail is a long, whip-like structure responsible for the motility of sperm. It is formed by microtubules arranged in a 9+2 axonemal structure. The movement of the tail propels the sperm forward, enabling it to reach and fertilize the ovum.

10. Write applications of dactylography.

APPLICATIONS OF DACTYLOGRAPHY:

- ***Criminal investigation (Forensics):*** Used to identify criminals by matching fingerprints found at crime scenes with fingerprint records, helping in solving crimes such as theft, murder, and assault.
- ***Civil identification:*** Used in national identity cards, passports, voter registration, and driving licenses to ensure accurate identification of individuals and prevent identity fraud.
- ***Biometric security systems:*** Used in mobile phones, laptops, ATMs, and banking systems for secure access and authentication based on fingerprint recognition.
- ***Government and administrative records:*** Used for maintaining employee attendance systems, pension verification, and public service records to ensure authenticity and prevent duplication.

11. Differentiate between Zona Pellucida, Corona Radiata and Follicular cells.

Feature	Zona Pellucida	Corona Radiata	Follicular Cells
Definition	Thick, transparent glycoprotein layer surrounding the oocyte membrane	Outer layer of follicle cells surrounding the zona pellucida	Granulosa (supporting) cells forming the ovarian follicle structure
Location	Lies just outside the plasma membrane of the ovum	Lies outside the zona pellucida around the egg	Found in ovarian follicle surrounding the developing oocyte
Structure	Acellular, non-cellular protective layer	Cellular layer of loosely arranged cells	Compact layer of somatic cells forming follicle wall
Composition	Glycoprotein matrix secreted by oocyte and follicular cells	Living follicular (granulosa) cells	Granulosa cells with supporting connective tissue
Function	Provides species-specific sperm binding and prevents polyspermy	Protects and nourishes oocyte; assists sperm penetration	Supports growth and nourishment of oocyte during development
Role in fertilization	Acts as a binding site for sperm and blocks polyspermy after fertilization	First barrier for sperm; dispersed by hyaluronidase enzyme	Helps in development of corona radiata and egg maturation
Nature	Non-cellular protective coat	Cellular protective layer around egg	Structural and nutritive ovarian cells

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12. Write functions of cuticle or bloom of eggshell.

Functions of cuticle (bloom) of eggshell:

- The cuticle is a thin, outermost organic coating on the eggshell that acts as a protective barrier against microorganisms**,** reducing the entry of bacteria and fungi through shell pores.
- It helps in reducing water loss by partially sealing the pores of the eggshell, thereby maintaining internal moisture balance.
- The cuticle provides mechanical protection to the shell surface and reduces surface damage during handling.
- It plays a role in maintaining egg freshness and quality by preventing contamination and slowing down deterioration.
- The bloom also gives a natural glossy appearance to freshly laid eggs and is often removed in washed commercial eggs.

13. What is the purpose of cortical granule reaction?

Purpose of cortical granule reaction:

The cortical granule reaction is a key post-fertilization event that occurs in the ovum immediately after the entry of the first sperm. Cortical granules present beneath the oocyte plasma membrane fuse with it and release their enzymes into the perivitelline space.

The main purpose of this reaction is to block polyspermy, ensuring that only one sperm fertilizes the egg. The released enzymes cause biochemical and structural modifications in the zona pellucida (or vitelline layer), leading to its hardening and formation of a fertilization membrane. This prevents any additional sperm from binding or penetrating the egg.

In addition, the cortical granule reaction helps in establishing the normal diploid condition (2n) of the zygote by avoiding abnormal chromosome numbers that would result from multiple sperm entry. It also contributes to creating a stable and protected environment for early embryonic development by strengthening the outer egg layer and stopping further sperm-egg interactions.

14. Write functions of Eggshell, chalaza, cuticle, air cell and vitelline membrane.

Eggshell: The eggshell is the hard, calcareous outer covering of the egg. Its main function is to provide mechanical protection to the developing embryo and prevent physical damage. It is porous in nature, allowing exchange of gases such as oxygen and carbon dioxide, which is essential for embryonic respiration.

Chalaza: The chalaza is a pair of twisted, rope-like structures made of albumen that anchor the yolk in the center of the egg. Its function is to maintain the correct position and orientation of the yolk and protect the embryo from mechanical shocks caused by movement of the egg.

Cuticle (bloom): The cuticle is the outermost thin protective layer of the eggshell. It functions as a barrier against microbial infection by blocking shell pores. It also reduces water loss and helps maintain egg freshness and internal moisture balance.

Air cell: The air cell is a small pocket of air present at the broader end of the egg between the shell membranes. It provides oxygen to the developing embryo, especially during the later stages of incubation, and also helps in cushioning the internal contents.

Vitelline membrane: The vitelline membrane surrounds the yolk and holds it intact. It maintains the structural integrity of the yolk and plays an important role in fertilization by allowing sperm binding and preventing the mixing of yolk with albumen.

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15. What is induced spawning?

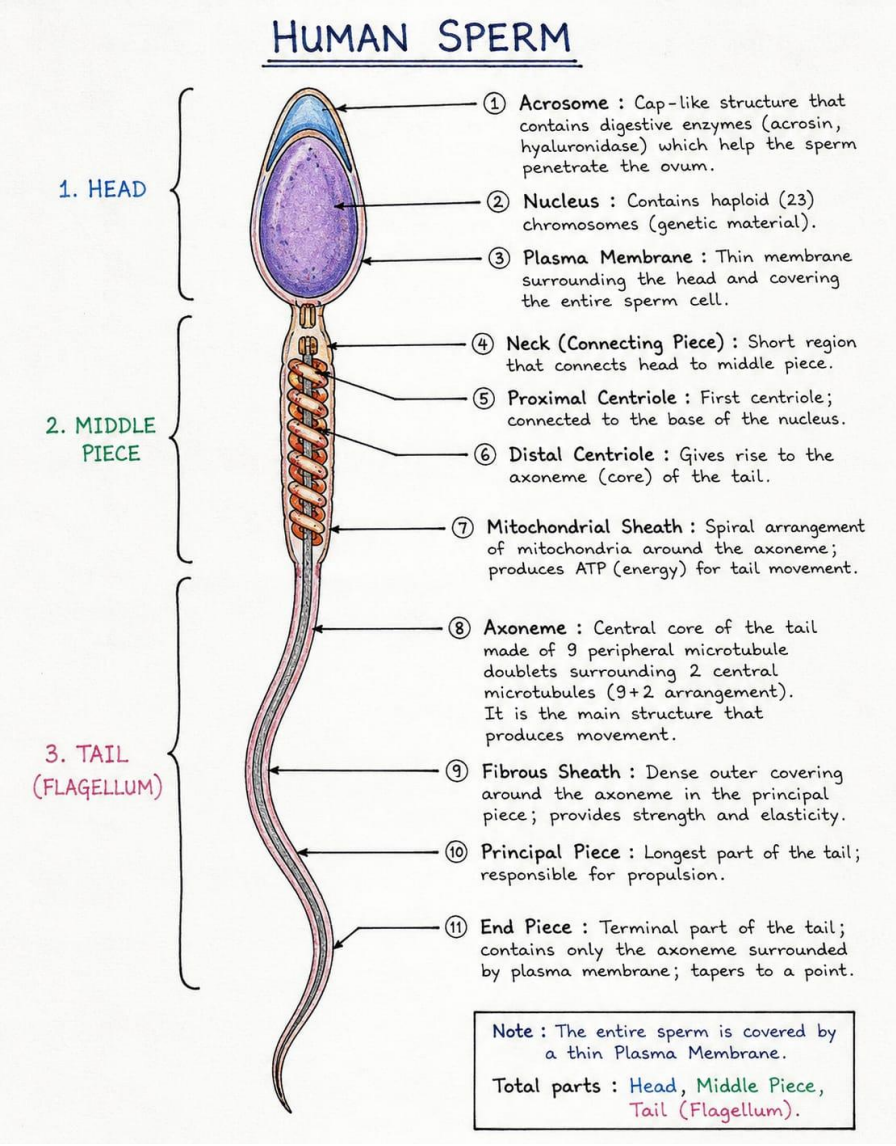
INDUCED SPAWNING:

Induced spawning is an artificial method used in aquaculture to stimulate fish to release eggs (ovulation) and sperm (spermiation) under controlled hatchery conditions. It is mainly applied to fish species that do not breed naturally in captivity due to absence of suitable environmental factors such as temperature, rainfall, water flow, or photoperiod.

In this technique, mature brood fish are selected and injected with hormonal preparations such as Ovaprim, pituitary gland extract, or other gonadotropin-releasing hormones. These hormones activate the endocrine system of fish, leading to final maturation of gametes, ovulation in females, and release of milt in males. After a specific latency period, the fish are either allowed to spawn naturally in tanks or eggs and milt are collected by stripping and then fertilization is done artificially.

Induced spawning is highly important in aquaculture because it ensures reliable and timely production of fish seed, overcomes seasonal breeding limitations, and allows large-scale production of economically important fish species for fish farming and conservation programs.

16. Label the given picture.



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